Target Profile Report #4

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CIS 4710 Information Security

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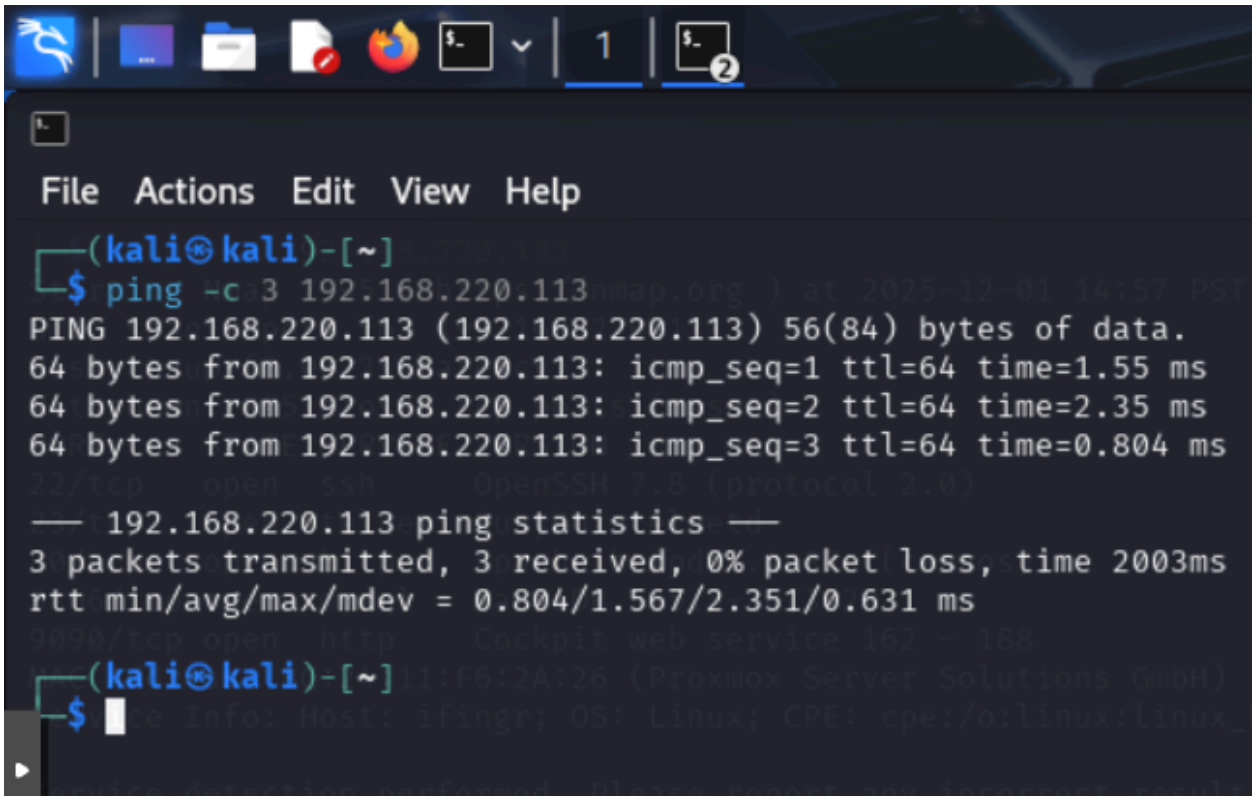
Target Selected

- **Target Name: 1453 (ohio.stream.stream)**
- **Target IP: 192.168.220.113**

Basic Connectivity

- **Ping Test: Ran ping -c 3 192.168.220.113 (ICMP replies received).**



Passive Reconnaissance

- **Reverse DNS Lookup Result: Both dig -x and nslookup commands failed with "no servers could be reached." The failed commands indicate that the 192.168.220.0/24 network does not provide reverse DNS resolution.**

- **Whois Lookup Result: The whois 192.168.220.113 command also failed due to lack of external DNS resolution.**

```
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ dig -x 192.168.220.113
;; communications error to 192.168.1.1#53: timed out
;; communications error to 192.168.1.1#53: timed out
;; communications error to 192.168.1.1#53: timed out

; <<>> DiG 9.20.4-4-Debian <<>> -x 192.168.220.113
;; global options: +cmd
;; no servers could be reached
┌──(kali㉿kali)-[~]
└─$ nslookup 192.168.220.113
;; communications error to 192.168.1.1#53: timed out
;; communications error to 192.168.1.1#53: timed out
;; communications error to 192.168.1.1#53: timed out
;; no servers could be reached

┌──(kali㉿kali)-[~]
└─$ whois 192.168.220.113
getaddrinfo(whois.arin.net): Temporary failure in name resolution
┌──(kali㉿kali)-[~]
└─$
```

- **TLS/SSL Certificate Analysis: Ran openssl s_client -connect 192.168.220.113:443 and got the following output:**
  - **Port/Service: 443/tcp — ssl/http (HTTPS)**
  - **Issuer: CN=Caddy Local Authority - ECC Intermediate**
  - **Subject: Not set**
  - **Validity: Dec 2 23:45:20 2025 GMT; Dec 3 11:45:20 2025 GMT**
  - **Public Key: ECC (Elliptic Curve), 256-bit**
  - **TLS Version / Cipher: TLSv1.3, Cipher = TLS_CHACHA20_POLY1305_SHA256**

kali@kali: ~

File Actions Edit View Help

```
read R BLOCK
^C
┌──(kali㉿kali)-[~]
└──(kali㉿kali)-[~]
└─$ openssl s_client -connect 192.168.220.113:443
Connecting to 192.168.220.113
CONNECTED(00000003)
Can't use SSL_get_servername
depth=1 CN=Caddy Local Authority - ECC Intermediate
verify error:num=20:unable to get local issuer certificate
verify return:1
depth=0
verify return:1
---
Certificate chain
 0 s:
   i:CN=Caddy Local Authority - ECC Intermediate
   a:PKEY: id-ecPublicKey, 256 (bit); sigalg: ecdsa-with-SHA256
   v:NotBefore: Dec  2 23:45:20 2025 GMT; NotAfter: Dec  3 11:45:20 2025 GMT
 1 s:CN=Caddy Local Authority - ECC Intermediate
   i:CN=Caddy Local Authority - 2023 ECC Root
   a:PKEY: id-ecPublicKey, 256 (bit); sigalg: ecdsa-with-SHA256
   v:NotBefore: Nov 30 22:53:09 2025 GMT; NotAfter: Dec  7 22:53:09 2025 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIBuTCCAV+gAwIBAgIRAJe0pkEocYP52uNhfGkXTzcwCgYIKoZIzj0EAwIwMzEx
MC8GA1UEAxMoQ2FkZHkgTG9jYWwgQXV0aG9yaXR5IC0gRUNDIEludGVybWVkaWF0
ZTAeFw0yNTEyMDIyMzQ1MjBaFw0yNTEyMDMxMTQ1MjBaMAAwWTATBgcqhkjOPQIB
BggqhkjOPQMBBwNCAASZivZ38S5BrWNEI56cNTJrVmU/9qpKLwYwh/EqJ/IKPVFg
03koxKCFZ28DuSJKGMwU+LpYLqM702EAXZRrWBGzo4GGMIGDMA4GA1UdDwEB/wQE
AwIHgDAdBgNVHSUEFjAUBggrBgEFBQcDAQYIKwYBBQUHAwIwHQYDVR0OBBYEFBQj
gydfnWl6u8fkXqhXSvUAZSZNMB8GA1UdIwQYMBaAFJiful20uyAcGpUGbzjEUbHa
8aQpMBIGA1UdEQEB/wQIMAaHBApZAAQwCgYIKoZIzj0EAwIDSAAwRQIhAM+5OTma
eb0hUXdQg9CSOz6JEzlvkFCqccksy5uvlJAsAiApGXHqvAQmyDN6gPpxtCKs4onN
2fC9daiVX7UJ2pCURg==
-----END CERTIFICATE-----
subject=
issuer=CN=Caddy Local Authority - ECC Intermediate
```

```
kali@kali: ~
File  Actions  Edit  View  Help
SSL handshake has read 1266 bytes and written 502 bytes
Verification error: unable to get local issuer certificate
New, TLSv1.3, Cipher is TLS_CHACHA20_POLY1305_SHA256
Protocol: TLSv1.3
Server public key is 256 bit
This TLS version forbids renegotiation.
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 20 (unable to get local issuer certificate)
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol  : TLSv1.3
    Cipher    : TLS_CHACHA20_POLY1305_SHA256
    Session-ID: 865D1A147C42177749D3A7E372DFFB6710CB7364E374013C279CA70E2E7BA08D
    Session-ID-ctx:
    Resumption PSK: 742534F001ACCF752A684FEFB0DC144E8AC7E30F660464FF10692952FD89CE79
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 604800 (seconds)
    TLS session ticket:
    0000 - fb fa ec c3 51 c7 30 7d-ca af bf 15 33 3f c9 0b   ....Q.0}....3?..
    0010 - 27 74 92 3b ac 92 08 f2-6b 1d da 36 50 a7 52 b2   't.;....k..6P.R.
    0020 - 8b 4d 5b 2b 26 52 91 d3-26 2d af 1a d4 21 d7 a9   .M[+&R..&-...!..
    0030 - b2 b0 31 7a c1 26 c9 1b-52 2e f2 ac 5d c0 5d 66   ..1z.&..R...].]f
    0040 - 3c 82 4e 79 aa 14 b1 37-c9 7a 06 4b 5c 3e ab f2   <.Ny...7.z.K\>..
    0050 - 7b cb bb 05 b1 55 5d f6-b4 7a af ab 31 96 f0 7a   {....U]..z..1..z
    0060 - c7 f8 3f 0b 57 8e 44 53-32                        ..?.W.DS2
    Start Time: 1764719563
    Timeout   : 7200 (sec)
    Verify return code: 20 (unable to get local issuer certificate)
    Extended master secret: no
    Max Early Data: 0
read R BLOCK
```

- **TLS/SSL Summary: This output shows the host is running a web server (Caddy) with HTTPS enabled and using secure encryption. The certificate is self-signed.**

Active Reconnaissance

```
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ nmap -p- 192.168.220.113
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-07 18:02 PST
Nmap scan report for 192.168.220.113
Host is up (0.00082s latency).
Not shown: 65525 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
139/tcp  open  netbios-ssn
443/tcp  open  https
445/tcp  open  microsoft-ds
4433/tcp open  vop
5355/tcp open  llmnr
5432/tcp open  postgresql
9090/tcp open  zeus-admin
MAC Address: BC:24:11:15:30:20 (Proxmox Server Solutions GmbH)

Nmap done: 1 IP address (1 host up) scanned in 17.01 seconds
```

Nmap -p- 192.168.220.113

-This scan enumerates all 65,535 TCP ports and identifies which ones are open on the target host. The results confirm the presence of several exposed services, indicating a broad attack surface.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV -sC -O 192.168.220.113
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-07 18:07 PST
Nmap scan report for 192.168.220.113
Host is up (0.0013s latency).
Not shown: 992 closed tcp ports (reset)
PORT     STATE SERVICE      VERSION
22/tcp   open  ssh          OpenSSH 8.9 (protocol 2.0)
| ssh-hostkey:
|   256 6e:90:d0:6c:b1:60:8d:b7:38:f6:db:34:e3:02:6e:f2 (ECDSA)
|_  256 c7:a3:4b:74:9b:17:87:17:b5:a4:82:6c:18:4b:fd:0f (ED25519)
53/tcp   open  domain       (generic dns response: NOTIMP)
| dns-nsid:
|   NSID: res120.bur (7265733132302e627572)
|   id.server: res120.bur
|_  bind.version: Q9-P-7.6
| fingerprint-strings:
|   DNSVersionBindReqTCP:
|     version
|     bind
|     version
|     bind
|_    Q9-P-7.6
80/tcp   open  http         Caddy httpd
|_http-title: /
|_http-server-header: Caddy
139/tcp  open  netbios-ssn  Samba smbd 4
443/tcp  open  ssl/http     Caddy httpd
|_http-server-header: Caddy
| ssl-cert: Subject:
| Subject Alternative Name: IP Address:10.89.0.4
| Not valid before: 2025-12-08T02:07:22
|_Not valid after:  2025-12-08T14:07:22
|_http-title: Site doesn't have a title.
445/tcp  open  netbios-ssn  Samba smbd 4
5432/tcp open  postgresql   PostgreSQL DB 15.2 - 15.4
9090/tcp open  ssl/zeus-admin?
| ssl-cert: Subject: commonName=ohio/organizationName=0ae550efe566428fb628cff80044a0fd
| Subject Alternative Name: IP Address:127.0.0.1, DNS:localhost
| Not valid before: 2025-11-08T04:32:32
|_Not valid after:  2026-12-08T04:32:32
```

```
|_http-server-header: Caddy
| ssl-cert: Subject:
| Subject Alternative Name: IP Address:10.89.0.4
| Not valid before: 2025-12-08T02:07:22
|_Not valid after:  2025-12-08T14:07:22
|_http-title: Site doesn't have a title.
445/tcp  open  netbios-ssn   Samba smbd 4
5432/tcp open  postgresql    PostgreSQL DB 15.2 - 15.4
9090/tcp open  ssl/zeus-admin?
| ssl-cert: Subject: commonName=ohio/organizationName=0ae550efe566428fb628cff80044a0fd
| Subject Alternative Name: IP Address:127.0.0.1, DNS:localhost
| Not valid before: 2025-11-08T04:32:32
|_Not valid after:  2026-12-08T04:32:32
|_ssl-date: TLS randomness does not represent time
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.c
gi?new-service :
SF-Port53-TCP:V=7.95%I=7%D=12/7%Time=693632F0%P=x86_64-pc-linux-gnu%r(DNSV
SF:ersionBindReqTCP,41,"\0\?\0\x06\x81\x80\0\x01\0\x01\0\0\0\0\x07version\
SF:x04bind\0\0\x10\0\x03\x07version\x04bind\0\0\x10\0\x03\0\0\0\x1e\0\t\x0
SF:8Q9-P-7\.6")%r(DNSStatusRequestTCP,E,"\0\x0c\0\0\x90\x04\0\0\0\0\0\0\
SF:0");
MAC Address: BC:24:11:15:30:20 (Proxmox Server Solutions GmbH)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop

Host script results:
|_nbstat: NetBIOS name: OHIO, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb2-time:
|   date: 2025-12-08T02:08:05
|_  start_date: N/A
|_clock-skew: -20s
| smb2-security-mode:
|   3:1:1:
|_    Message signing enabled but not required

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 94.47 seconds

┌──(kali㉿kali)-[~]
└─$
```

nmap -sV -sC -O 192.168.220.113

- This scan collects detailed service versions and runs default Nmap scripts to gather additional metadata such as SSL certificates, SMB information, and OS fingerprints. It reveals the host is likely a Proxmox-managed Linux system running multiple services including Caddy, Samba, PostgreSQL, and DNS.

```
┌──(kali㉿kali)-[~]
└─$ curl -I http://192.168.220.113
HTTP/1.1 200 OK
Content-Type: text/html; charset=utf-8
Server: Caddy
Date: Mon, 08 Dec 2025 02:10:25 GMT
```

Curl -I http://192.168.220.113

- The curl header request retrieved HTTP response headers, confirming that the web server is running Caddy and returning a standard HTTP 200 OK response. This verifies proper web service operation on port 80.

```
┌──(kali㉿kali)-[~]
└─$ whatweb 192.168.220.113
http://192.168.220.113 [200 OK] Country[RESERVED][ZZ], HTML5, HTTPServer[Caddy], IP[192.168.220.113], Script, Title[/]
```

Whatweb 192.168.220.113

- This scan utilized WhatWeb to identify the target as hosting HTML5 content served by the Caddy web server and providing metadata such as country code and IP. The minimal fingerprint suggests a simple or default web deployment without frameworks like WordPress or Django.

```
┌──(kali㉿kali)-[~]
└─$ nc 192.168.220.113 80
HTTP/1.1 400 Bad Request
Content-Type: text/plain; charset=utf-8
Connection: close

400 Bad Request
```

Nc 192.168.220.113

- This scan utilized netcat to connect to port 80 shows how the server responds to malformed or raw input. The 400 Bad Request message indicates the HTTP service is active and requires valid HTTP formatting to proceed.